GRADE 7 ACCELERATED · UNIT 5

Unit 5 Review

Multi-Step Proportional Relationships **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers every learning target from the unit — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Scale Drawings

Lesson 1 · MA.7.GR.1.5 · 5 questions

1. A 5 cm side of a figure is 8 cm in its scale copy. Find the scale factor.

k = 1.6

copy ÷ original

2. A 16 cm side of a figure is 4 cm in its scale copy. Find the scale factor.

k = 0.25

the copy is smaller, so k is under 1

3. A scale copy is made at $k = 1.5$. A side of the original is 6 cm. Find the matching side of the copy.

9 cm

4. A drawing has a 65° angle. Its scale copy is made at $k = 2$. What is the matching angle in the copy?

- ○ A. 32.5°
- ○ B. 67°
- ○ C. 130°
- ● **D. 65°**

D

5. A map is drawn at $1\,\text{cm} : 30\,\text{km}$. Two towns are 4.5 cm apart on it. How far apart are they?

135 km

drawing × rate

PART B. Scale Factor: Perimeter and Area

Lessons 2–3 · MA.7.GR.1.5 · 6 questions

6. A figure with perimeter 16 cm is scaled by $k = 2.5$. Find the copy's perimeter.

40 cm

perimeter × k

7. A figure with area $9\,\text{cm}^2$ is scaled by $k = 2.5$. Find the copy's area.

56.25 cm²

area × k^2 = 9 × 6.25

8. A 4 cm by 6 cm rectangle is scaled by $k = \frac{1}{2}$.

a. Find the copy's perimeter.

10 cm

20 × ½

b. Find the copy's area.

6 cm²

24 × ¼

9. A figure has area $10\,\text{cm}^2$ and its scale copy has area $160\,\text{cm}^2$. Find k.

k = 4

160 ÷ 10 = 16 = k^2

10. A figure's perimeter is 12 cm and its copy's perimeter is 36 cm. What is the area ratio, copy to original?

9

k = 3 from the perimeters; the area ratio is k^2

11. Sam says tripling every side of a rectangle triples its area. Explain why not, with a counterexample.

A 2 by 5 (area 10) tripled is 6 by 15 (area 90): nine times, because area is two lengths and each is tripled.

the reason and the numbers

PART C. Real-World Scale Drawings

Lesson 4 · MA.7.GR.1.5 · 4 questions

12. A plan is drawn at $\frac{1}{2}\,\text{in} : 1\,\text{ft}$. A wall is 7.5 in long on it. Find the real length.

15 ft

1 in : 2 ft; 7.5 × 2

13. A plan is drawn at $1\,\text{in} : 6\,\text{ft}$. A real wall is 21 ft long. Find its length on the plan.

3.5 in

real ÷ rate

14. A model boat is built at $1 : 25$ and is 32 cm long. Find the real boat's length in meters.

8 m

32 × 25 = 800 cm

15. SSDD. Every part below is about one floor plan drawn at ¼ in : 1 ft, on which a room is 3 in by 4 in. 1 ft = 0.305 m. Same surface, four different questions: a length, an area with the rate used twice, the scale factor as one number, and a length conversion — the four moves of the unit on one room, so that a student learns to read the question and not the picture.

a. Find the real length of the wall drawn 3 in long.

12 ft

1 in : 4 ft; 3 × 4

b. Find the room's real area.

192 ft²

12 by 16; the rate used twice

c. Write the scale factor from the drawing to the room as a single number.

48

1 in : 4 ft is 1 in : 48 in

d. Find the room's real perimeter in meters.

17.08 m

2(12 + 16) = 56 ft; 56 × 0.305 (56 ÷ 3.28 = 17.07 by the other ratio). The area in square meters is the test's transfer item and is not rehearsed here (ruling 18).

PART D. Converting Length Across Systems; Currency *Lessons 6–7 · MA.7.AR.3.3 · 6 questions*

16. $1\text{ in} = 2.54\text{ cm}$. How many centimeters is 14 in?

35.56 cm

17. $1\text{ cm} = 0.39\text{ in}$. How many inches is 120 cm?

46.8 in (47.2 by ÷ 2.54)

fewer inches than centimeters

18. $1\text{ km} = 0.62\text{ mi}$. How many miles is 30 km?

18.6 mi

19. $1\text{ yd} = 3\text{ ft}$, $1\text{ ft} = 12\text{ in}$, $1\text{ in} = 2.54\text{ cm}$. How many centimeters is 6 yd? Estimate first.

About 540; exactly 548.64 cm

216 in × 2.54

20. Rate on Nov 3: $1\text{ US dollar} = 0.92\text{ euros}$.

a. How many euros is $150?

138 euros

× the rate

b. How many US dollars is 69 euros?

$75

÷ the rate

21. $1\text{ in} = 2.54\text{ cm}$; $1\text{ cm} = 0.39\text{ in}$. Lee converts 40 cm to inches by multiplying by 2.54 and gets 101.6 in. Name the error and fix it.

Lee used the ratio whose 1 is inches on centimeters. Inches are bigger, so there are fewer: 40 × 0.39 = 15.6 in (or 40 ÷ 2.54 = 15.7).

the error named and the estimate

PART E. Converting Mass, Area and Volume

Lessons 8–9 · MA.7.AR.3.3 · 5 questions

22. $1\text{ lb} = 0.454\text{ kg}$. How many kilograms is 80 lb?

36.32 kg

(80 ÷ 2.204 = 36.30 by the other ratio)

23. $1\text{ L} = 0.264\text{ gal}$. How many gallons is 40 L?

10.56 gal

24. $1\text{ ft} = 0.305\text{ m}$. How many square meters is 8 ft^2 ? (Round to two places.)

About 0.74 m^2

$8 \times 0.305^2 = 0.7442$: the sheet's length ratio, squared

25. $1\text{ c} = 8\text{ fl oz}$ and $1\text{ c} = 0.24\text{ L}$. How many liters is 48 fl oz?

1.44 L

6 c × 0.24

26. $1\text{ m} = 3.28\text{ ft}$. A desk is 1.5 m^2 in area. Which is closest to its area in square feet?

- ○ A. 4.9 ft^2
- ○ B. 9.8 ft^2
- ● **C.** 16.1 ft^2
- ○ D. 10.8 ft^2

C

$1.5 \times 3.28^2 = 16.1376$.

PART F. Multi-Step Proportion Problems

Lessons 10–11 · MA.7.AR.4.5 · 4 questions

27. A car gets 29 miles per gallon and its tank holds 12 gal.

a. How far does it go on a full tank?

348 mi

b. It starts an 800-mi trip with a full tank. How many tanks of gas does the trip take?

3 tanks (800 ÷ 348 ≈ 2.3)

a part of a tank is still a tank

28. A recipe for 3 servings uses $1\frac{1}{4}$ cups of rice.

a. How much rice for 12 servings?

5 cups

k = 4

b. With 10 cups on hand, how many servings can be made?

24 servings

8 recipes × 3

29. $1\,\text{km} = 0.62\,\text{mi}$. A train does 110 km/h; a bus does 65 mi/h. Which is faster? Show both speeds in one unit.

The train: 110 km/h = 68.2 mi/h > 65 mi/h

same unit first

30. Paint is mixed $1:2:5$. A batch uses 10 gal of the third color. How many gallons is the whole batch?

16 gal

k = 2; 8 parts × 2